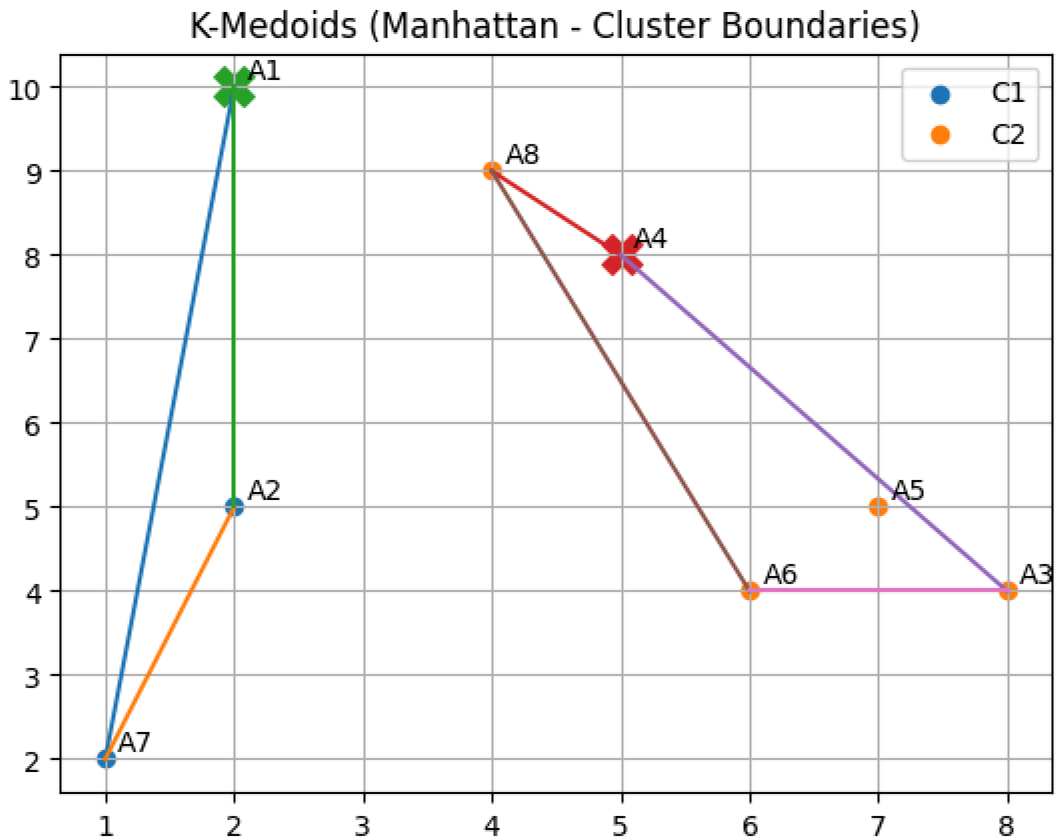
DATA SCIENCE MID 2 SOLUTION

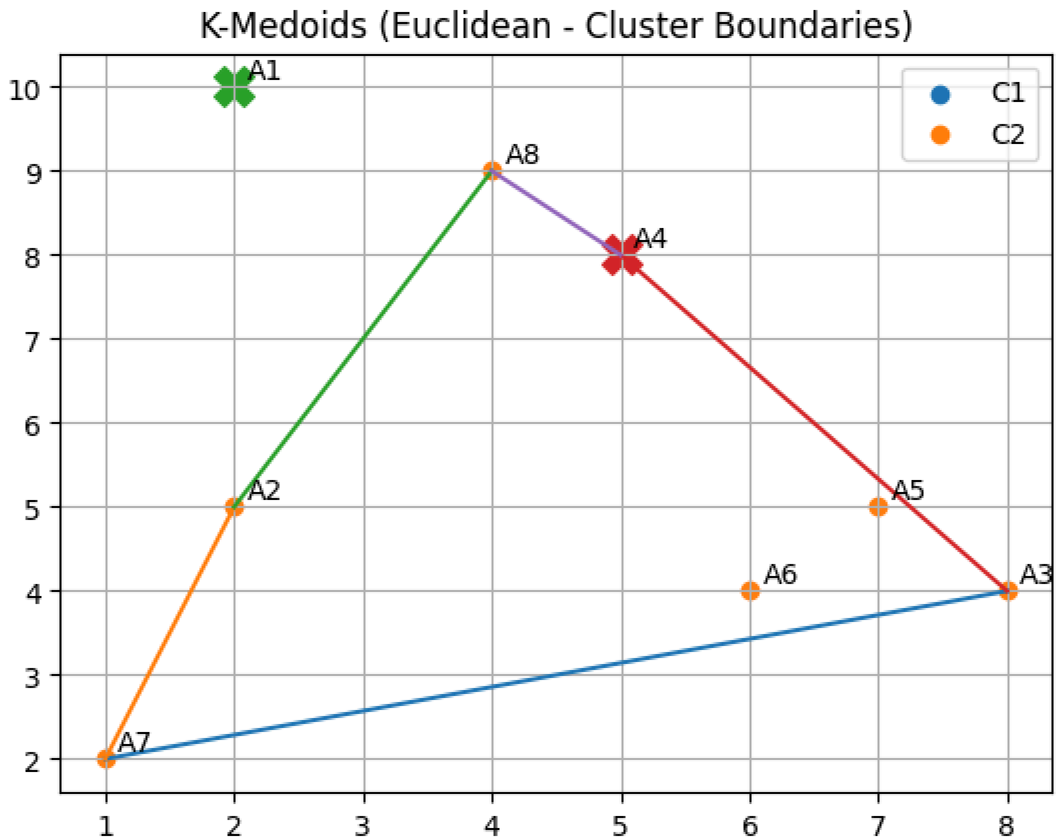
Euclid

	$C_1 = (2,10)$	$C_2 = (5,8)$	Cluster Assignment.
A_1 2,10	$\sqrt{(2-2)^2+(10-10)^2}=0$	$\sqrt{(5-2)^2+(8-10)^2}=3.60$	C_1
A_2 2,5	$\sqrt{(2-2)^2+(10-5)^2}=5$	$\sqrt{(5-2)^2+(8-5)^2}=4.24$	C_2
A_3 8,4	$\sqrt{(2-8)^2+(10-4)^2}=8.48$	$\sqrt{(5-8)^2+(8-4)^2}=5$	C_2
A_4 5,8	$\sqrt{(2-5)^2+(10-8)^2}=3.60$	$\sqrt{(5-5)^2+(8-8)^2}=0$	C_2
A_5 7,5	$\sqrt{(2-7)^2+(10-5)^2}=7.07$	$\sqrt{(5-7)^2+(8-5)^2}=3.60$	C_2
A_6 6,4	$\sqrt{(2-6)^2+(10-4)^2}=7.21$	$\sqrt{(5-6)^2+(8-4)^2}=4.123$	C_2
A_7 1,2	$\sqrt{(2-1)^2+(10-2)^2}=8.06$	$\sqrt{(5-1)^2+(8-2)^2}=7.21$	C_2
A_8 4,9	$\sqrt{(2-4)^2+(10-9)^2}=2.23$	$\sqrt{(5-4)^2+(8-9)^2}=1.414$	C_2

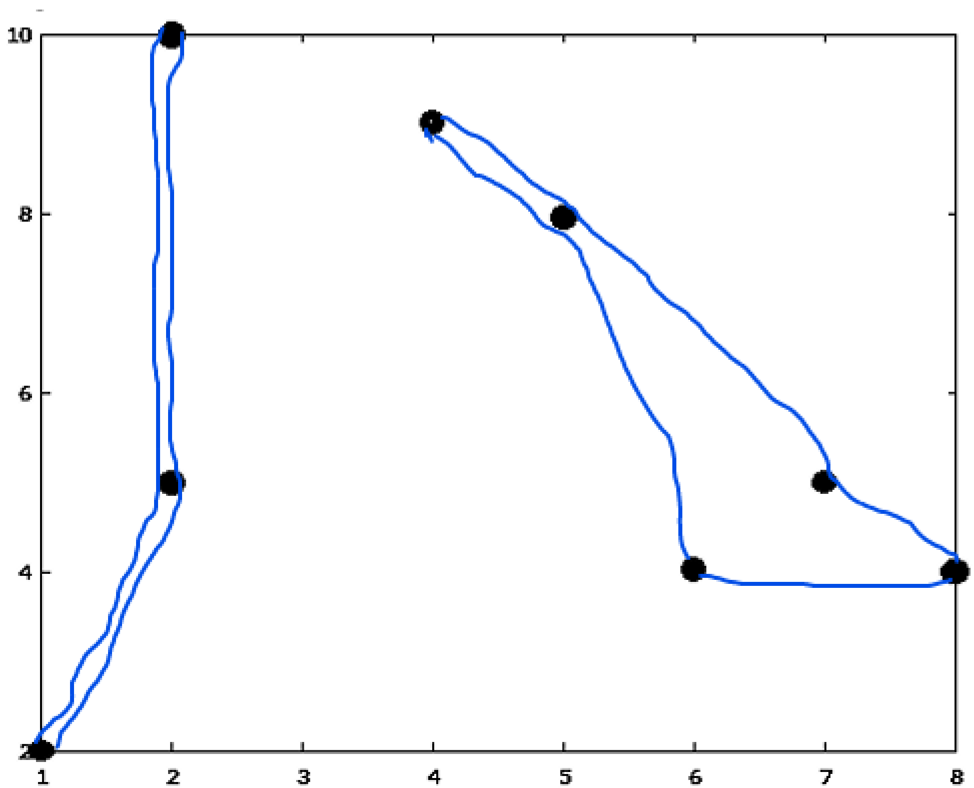
Manhattan.

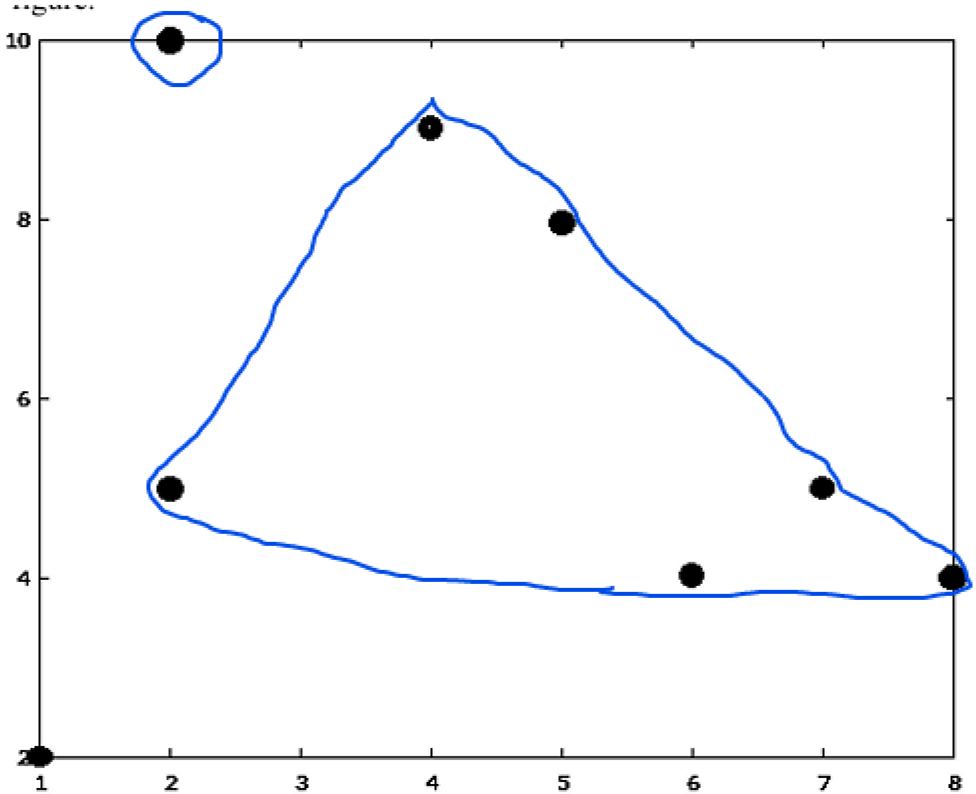
	$C_1 = 2,10$	$C_2 = (5,8)$	Cluster Assignm
A_1 2,10	$\lvert 2-2\rvert+\lvert 10-10\rvert=0$	$\lvert 5-2\rvert+\lvert 8-10\rvert=5$	C_1
A_2 2,5	$\lvert 2-2\rvert+\lvert 10-5\rvert=5$	$\lvert 5-2\rvert+\lvert 8-5\rvert=6$	C_1
A_3 8,4	$\lvert 2-8\rvert+\lvert 10-4\rvert=12$	$\lvert 5-8\rvert+\lvert 8-4\rvert=7$	C_2
A_4 5,8	$\lvert 2-5\rvert+\lvert 10-8\rvert=5$	$\lvert 5-5\rvert+\lvert 8-8\rvert=0$	C_2
A_5 7,5	$\lvert 2-7\rvert+\lvert 10-5\rvert=10$	$\lvert 5-7\rvert+\lvert 8-5\rvert=5$	C_2
A_6 6,4	$\lvert 2-6\rvert+\lvert 10-4\rvert=10$	$\lvert 5-6\rvert+\lvert 8-4\rvert=5$	C_2
A_7 1,2	$\lvert 2-1\rvert+\lvert 10-2\rvert=9$	$\lvert 5-1\rvert+\lvert 8-2\rvert=10$	C_1
A_8 4,9	$\lvert 2-4\rvert+\lvert 10-9\rvert=3$	$\lvert 5-4\rvert+\lvert 8-9\rvert=2$	C_2


K-Medoids (Manhattan - Cluster Boundaries)
A1
A2
A3
A4
A5
A6
A7
A8
C1
C2
1
2
3
4
5
6
7
8
9
10


K-Medoids (Euclidean - Cluster Boundaries)
C1
C2
A1
A8
A4
A2
A5
A6
A3
A7
10
9
8
7
6
5
4
3
2
1
2
3
4
5
6
7
8

DATA SCIENCE MID 2 SOLUTION





DATA SCIENCE MID 2 SOLUTION

TID	Items
1	A1, A5, A6, A8
2	A2, A4, A8
3	A4, A5, A7
4	A2, A3
5	A5, A6, A7
6	A2, A3, A4
7	A2, A6, A7, A9
8	A5
9	A8
10	A3 A5 A7
11	A3 A5 A7
12	A5 A6 A8
13	A2 A4 A6 A7
14	A1 A3 A5 A7
15	A2 A3 A9

Min Support = 3 So

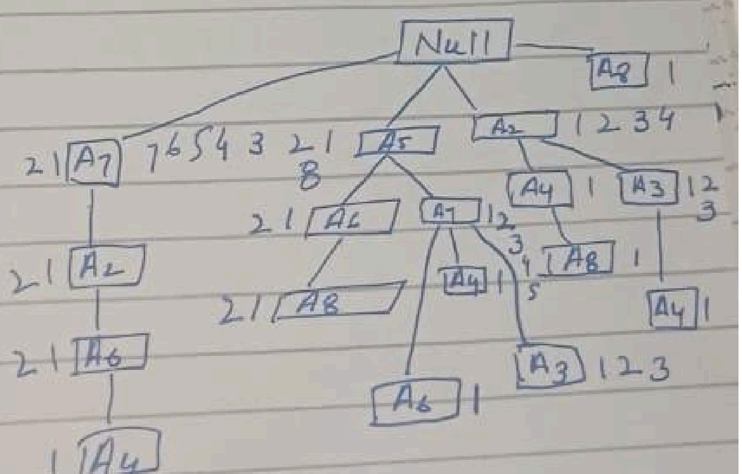
ItemSet	Support	
A1	2	X
A2	6	3
A3	6	4
A4	4	6
A5	8	1
A6	5	5
A7	7	2
A8	4	7
A9	2	X

Sort Frequency items in descending order based on support.

Itemset	Support
A5	8
A7	7
A2	6
A3	6
A6	5
A4	4
A8	4

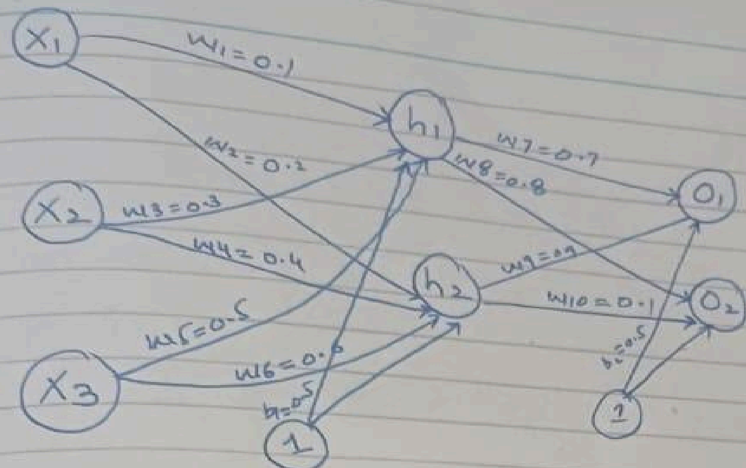
Sort Transactions (ordered Item set)

TID	Items
1	A5, A6, A8
2	A2, A4, A8
3	A5, A7, A4
4	A2, A3
5	A5, A7, A6
6	A2, A3, A4
7	A7, A2 A6
8	A5
9	A8
10	A5 A7 A3
11	A5 A7 A3
12	A5 A6 A8
13	A7 A2 A6 A4
14	A5 A7 A3
15	A2 A3



Item	CPB	Cond FP Tree
A_8	$\{\{A_5, A_6 : 2\}, \{A_2, A_4 : 1\}\}$	ϕ
A_4	$\{\{A_5, A_7 : 1\}, \{A_2, A_4 : 1\}, \{A_2, A_3 : 1\} \{A_7, A_2, A_6 : 1\}\}$	$\{A_2 : 3\}$
A_6	$\{\{A_5 : 2\} \{A_5, A_7 : 1\} \{A_7, A_2 : 2\}\}$	$\{A_5 : 3\} \{A_7 : 3\}$
A_3	$\{A_5, A_7 : 3\} \{A_2 : 3\}$	$\{A_5, A_7 : 3\} \{A_2 : 3\}$
A_2	$\{A_7 : 2\} \{A_5, A_7 : 1\}$	$\{A_7 : 3\}$
A_7	$\{A_5 : 5\}$	$\{A_5 : 5\}$

Item	Frequent Pattern Generated.
A_8	$\{\ \}$
A_4	$\{A_2, A_4 : 3\}$
A_6	$\{A_5\ A_6 : 3\} \{A_7, A_6 : 3\}$
A_3	$\{A_5, A_3 : 3\}, \{A_7, A_3 : 3\} \{A_5, A_7, A_3 : 3\} \{A_2, A_3 : 3\}$
A_2	$\{A_7, A_2 : 3\}$
A_7	$\{A_5, A_7 : 5\}$



$$h_1 = (x_1 * w_1) + (x_2 * w_3) + (x_3 * w_5) + b_1$$
$$= (1 \times 0.1) + (4 \times 0.3) + (5 \times 0.5) + 0.5$$
$$= 4.3$$
$$\hat{h}_1 = \frac{1}{1+e^{-4.3}} = \boxed{0.986}$$

$$h_2 = (x_1 * w_2) + (x_2 * w_4) + (x_3 * w_6) + b_2$$
$$= (1 \times 0.2) + (4 \times 0.4) + (5 * 0.6) + 0.5$$
$$= 5.3$$
$$\hat{h}_2 = \frac{1}{1+e^{-5.3}} = \boxed{0.995}$$

$$O_1 = (\hat{h}_1 * w_7) + (\hat{h}_2 * w_9) + b_{O1}$$
$$= (0.986 \times 0.7) + (0.995 \times 0.9) + 0.5$$
$$= 2.0857$$
$$\hat{O}_1 = \frac{1}{1+e^{-2.0857}} = \boxed{0.889}$$

$$O_2 = (\hat{h}_1 * w_8) + (\hat{h}_2 * w_{10}) + b_{O2}$$
$$(0.986 \times 0.8) + (0.995 * 0.1) + 0.5 = 1.388$$
$$\hat{O}_2 = \frac{1}{1+e^{-1.388}} = \boxed{0.800}$$

Loss

$$\frac{1}{2n}\Sigma(y-\hat{y})^2$$

$$\frac{1}{2}\Sigma(0.1-0.889)^2+(0.05-0.800)^2$$

$$\frac{1}{2}\Sigma(-0.789)^2+(-0.75)^2$$

$$\frac{1}{2}\Sigma(0.622+0.5625)$$

$$\frac{1.1845}{2}=\boxed{0.592}$$

$$W_7New = w_7old - \eta\left(\frac{\partial L}{\partial w_7}\right)$$

$$w_8New = w_8old - \eta\left(\frac{\partial L}{\partial w_8}\right)$$

$$w_9New = w_9old - \eta\left(\frac{\partial L}{\partial w_9}\right)$$

$$w_{10}New = w_{10}old - \eta\left(\frac{\partial L}{\partial w_{10}}\right)$$

$$b_0new = b_0old - \eta\left(\frac{\partial L}{\partial b_0}\right)$$

$$\frac{\partial L}{\partial w_7} = \frac{\partial L}{\partial \hat{O}_1}\times\frac{\partial \hat{O}_1}{\partial O_1}\times\frac{\partial O_1}{\partial w_7}$$

$$(\hat{O}_1 - O_1actual) * \hat{O}_1(1-\hat{O}) \times \hat{h}_1$$

$$= (0.889-0.1) * (0.889)(1-0.889) * 0.988$$

$$= 0.789 * 0.0986 \times 0.988$$

$$= 0.076$$

$$w_7New = w_7old - \eta\left(\frac{\partial L}{\partial w_7}\right)$$

$$w_7New = 0.7 - 0.01(0.076)$$

$$= 0.699$$

$$\frac{\partial L}{\partial w_8} = \frac{\partial L}{\partial \hat{O}_2} \times \frac{\partial \hat{O}_2}{\partial O_2} \times \frac{\partial O_2}{\partial w_8}$$

$$= (\hat{O}_2 - O_2 actual) * (\hat{O}_2(1-\hat{O}_2)) * \hat{h}_1$$

$$= (0.800 - 0.05) * 0.800(1-0.800) * 0.986$$

$$= 0.75 * 0.16 * 0.986$$

$$= 0.11832$$

$$w_8New = w_8old - \eta\left(\frac{\partial L}{\partial w_8}\right)$$

$$= 0.8 - 0.01(0.11832)$$

$$= 0.7988$$

$$\frac{\partial L}{\partial w_9} = \frac{\partial L}{\partial \hat{O}_1} \times \frac{\partial \hat{O}_1}{\partial O_1} \times \frac{\partial O_1}{\partial w_9}$$

$$= (\hat{O}_1 - O_1 actual) * \hat{O}_1(1-\hat{O}_1) * \hat{h}_2$$

$$= (0.889 - 0.1) * 0.889(1-0.889) * 0.995$$

$$= 0.789 * 0.0986 * 0.995$$

$$= 0.0774$$

$$w_9New = w_9old - \eta\left(\frac{\partial L}{\partial w_9}\right)$$

$$0.9 - 0.01(0.0774) = 0.8992$$

$$\frac{\partial L}{\partial w_{10}} = \frac{\partial L}{\partial \hat{O}_2} \times \frac{\partial \hat{O}_2}{\partial O_2} \times \frac{\partial O_2}{\partial w_{10}}$$

$$= (\hat{O}_2 - O_2 actual) * \hat{O}_2(1-\hat{O}_2) * \hat{h}_2$$

$$= (0.800 - 0.05) * 0.800(1-0.800) * 0.995$$

$$= 0.75 * 0.16 \times 0.995 = 0.1194$$

$$w_{10}New = w_{10}old - \eta\left(\frac{\partial L}{\partial w_{10}}\right)$$

$$w_{10}New = 0.1 - 0.01(0.1194)$$

$$w_{10}New = 0.0988$$

$$\frac{\partial L}{\partial b_{01}} = \frac{\partial L}{\partial \hat{O}_1} \times \frac{\partial \hat{O}_1}{\partial O_1} \times \frac{\partial O_1}{\partial b_{01}}$$

$$(\hat{O}_1 - O_1 actual) * \hat{O}_1(1-\hat{O}_1) * 1$$

$$= (0.889 - 0.1) * 0.889(1-0.889) \times 1$$

$$= 0.789 * 0.098679 \times 1$$

$$= 0.0778$$

$$b_{01}New = b_{01}old - \eta\left(\frac{\partial L}{\partial b_{01}}\right)$$

$$0.5 - 0.01(0.0778)$$

$$b_{01}New = \boxed{0.499}$$

$$w_{1new} = w_{1old} - \eta\left(\frac{\partial L}{\partial w_1}\right)$$

$$\frac{\partial L}{\partial w_1} = \left[\frac{\partial L}{\partial \hat{o}_1} \times \frac{\partial \hat{o}_1}{\partial o_1} \times \frac{\partial o_1}{\partial \hat{h}_1}\right] + \left[\frac{\partial L}{\partial \hat{o}_2} \times \frac{\partial \hat{o}_2}{\partial o_2} \times \frac{\partial o_2}{\partial \hat{h}_1}\right]$$

$$\frac{\partial \hat{h}_1}{\partial h_1} \times \frac{\partial h_1}{\partial w_1}$$

$$\frac{\partial L}{\partial w_1} = \Big[\big[(\hat{o}_1 - o_1\,actual) * (\hat{o}_1)(1-\hat{o}_1) * (w_7)\big] +$$

$$\big[(\hat{o}_2 - o_2\,actual) * \hat{o}_2(1-\hat{o}_2) * w_8\big]\Big] *$$

$$\hat{h}_1(1-\hat{h}_1) * x_1$$

$$= \Big[\big[(0.889 - 0.1) * 0.889(1-0.889) * 0.7\big] +$$

$$\big[(0.800 - 0.05) * 0.800(1-0.800) * 0.8\big]\Big] *$$

$$0.986(1-0.986) * 1$$

$$= [0.789 * 0.0986 \times 0.7] +$$

$$[0.75 * 0.16 \times 1] * 0.0138$$

$$[0.0544 + 0.12] \times 0.0138$$

$$0.1744 \times 0.0138$$

$$= 0.00240672$$

$$w_{1new} = w_{1old} - \eta\left(\frac{\partial L}{\partial w_1}\right)$$

$$0.1 - 0.01(0.00240672)$$

$$w_{1new} = \boxed{0.0999}$$

$$w_3New = w_3old - \eta\left(\frac{\partial L}{\partial w_3}\right)$$

$$\frac{\partial L}{\partial w_3} = \left[\frac{\partial L}{\partial \hat{o}_1} \times \frac{\partial \hat{o}_1}{\partial o_1} \times \frac{\partial o_1}{\partial \hat{h}_1}\right] + \left[\frac{\partial L}{\partial \hat{o}_2} \times \frac{\partial \hat{o}_2}{\partial o_2} \times \frac{\partial o_2}{\hat{h}_1}\right]$$

$$* \frac{\partial \hat{h}_1}{\partial h} \times \frac{\partial h_1}{\partial w_3}$$

$$0.1744 \times 0.01388 * 4$$

$$= 0.009682$$

$$w_3New = 0.3 - 0.01(0.009682)$$

$$= \boxed{0.299}$$

$$w_5New = 0.1744 \times 0.01388 * 5$$

$$= 0.0121$$

$$w_5New = 0.5 - 0.01(0.0121)$$

$$= \boxed{0.499}$$

$$\frac{\partial L}{\partial w_2} = \left[\frac{\partial L}{\partial \hat{o}_1} \times \frac{\partial \hat{o}_1}{\partial o_1} \times \frac{\partial o_1}{\hat{h}_2}\right] + \left[\frac{\partial L}{\partial \hat{o}_2} \times \frac{\partial \hat{o}_2}{\partial o_2} \times \frac{\partial o_2}{\partial \hat{h}_2}\right]$$

$$* \frac{\partial \hat{h}_2}{\partial h_2} \times \frac{\partial h_2}{\partial w_2}$$

$$= \left[(\hat{o}_1 - o_{1\,actual}) * \hat{o}_1(1-\hat{o}_1) * w_9\right] +$$

$$\left[(\hat{o}_2 - o_{2\,actual}) * \hat{o}_2(1-\hat{o}_2) * w_{10}\right] *$$

$$\hat{h}_2(1-\hat{h}_2) * x_1$$

$$= [(0.889 - 0.1) * 0.889(1-0.889) * 0.9] +$$

$$[(0.800 - 0.05) * 0.800(1-0.800) * 0.1] +$$

$$0.995(1-0.995) * 1$$

$$([0.789 * 0.098679 \times 0.9] +$$

$$[0.75 * 0.16 * 0.1]$$

$$* 0.004975)$$

$$= [0.0700 + 0.012] * 0.004975$$

$$0.082 * 0.004975 = 0.0004079$$

$$w_{2new} = 0.2 - 0.01(0.0004079) = \boxed{0.1999}$$

$$w_{4new} = 0.4 - 0.01(0.082 \times (0.004975 \times 4)) = \boxed{0.399}$$

$$= (0.00163)$$

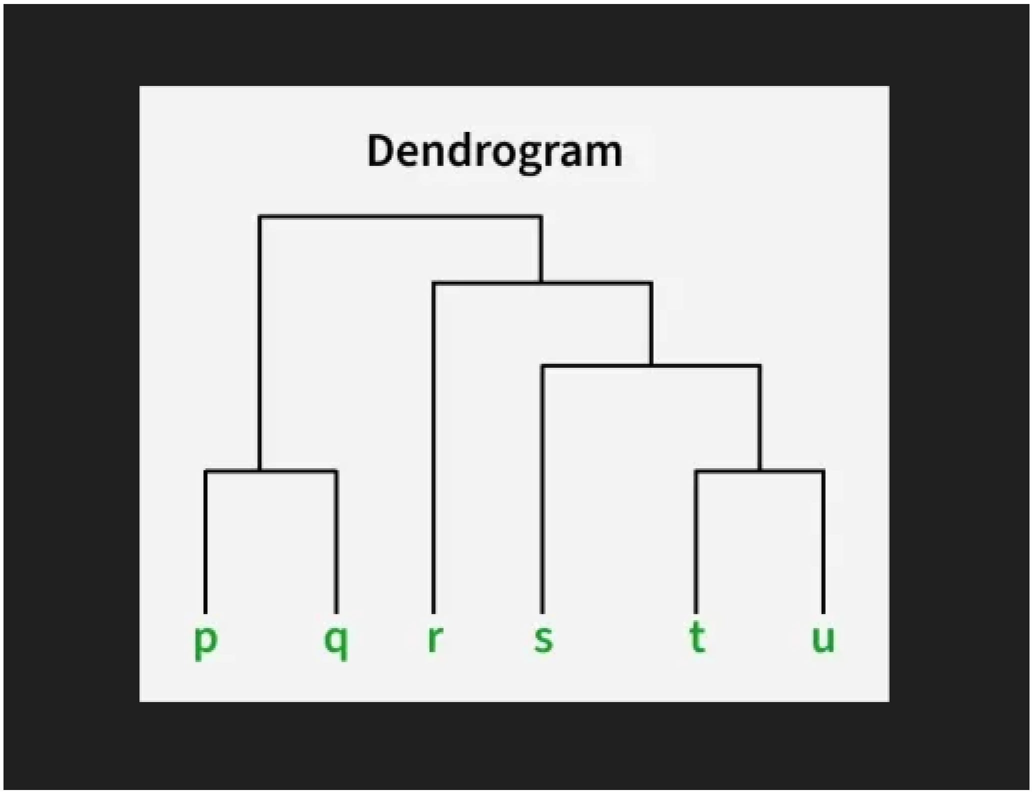
$$w_{6new} = 0.6 - 0.01(0.082 \times (0.004975 \times 6)) = \boxed{0.5999}$$

$$= 0.0024477$$

$$b_{hnew} = 0.5 - 0.01(0.082 \times 0.004975 \times 1) = \boxed{0.499}$$

DATA SCIENCE MID 2 SOLUTION

Answer



ELBOW METHOD

SILHOUTE ANALYSIS

Dendograms are exclusively used in Hierarchical Clustering (Agglomerative or Divisive).

Visualizing the Hierarchy of Data

No Need for Initial Centroids

K-means is sensitive to how you initialize the centroids (random seeds). Dendrograms (specifically Agglomerative clustering) are deterministic; given the same data and distance metric, you will always get the exact same tree.